

# Taichi Kamei

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Embedded systems & Hardware engineer who takes projects from 0 to 1 under real physical constraints

## EDUCATION

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### 4th Year Engineering Physics

*Expected Graduation - May 2028*

University of British Columbia | Vancouver

Relevant Courses: Solid and Fluid Mechanics, Mechanical Designs, Electric Circuit Analysis, Signals and Systems, Software Construction, Digital Systems and Microcomputers, Statistical Mechanics, Quantum Mechanics

## EXPERIENCE

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### Firmware & Hardware Engineer Co-op

*May 2026 - Present*

UBC Blusson Quantum Matter Institute

- Initially started as a PCB validation role for the 4-channel voltage controlled biasing current source PCB used for the superconducting nanowire single photon detector (SNSPD)
- Recognized the need for firmware development, designed an active object architecture in C/C++ on ESP32 with ESP-IDF, FreeRTOS, dual-core task isolation, inter-core queue communication, and event coordination, achieving deterministic concurrent per-channel control
- Ported Analog Devices no-OS 24-bit Sigma-Delta ADC and 16-bit DAC drivers to ESP-IDF, implemented SPI platform abstraction layer with manual CS toggle for 5 SPI slaves and mixed SPI modes on a 10MHz shared bus
- Diagnosed ADC garbage reading due to SPI CPOL mismatch on different SPI modes (DAC: Mode 1, ADC: Mode 3), resolved the issue by adding an NOP transaction before actual transaction
- Developed hysteresis V-I sweep mode with 10mV step size and steady-state biasing mode with 10nA precision
- Implemented calibration mode using Non-Volatile Storage, enabling runtime calibration lasting across power-cycle
- Debugged 3.3V LDO undervoltage by oscilloscope scoping and datasheet comparison, applied hardware workaround and PCB layout revision on Altium
- Tested time domain DAC and ADC drift characteristics by logging both PCB and commercial SMU measurement for 2 days using Raspberry Pi
- Designing a RF coplanar waveguide PCB and a low-pass filter PCB for GHz signals inside 2 Kelvin cryostats

## PROJECTS

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### Drone Flight Controller

*Feb 2026 - Present*

- Designed a power board for 4S LiPo, integrating BMS IC and 5V/5A buck converter
- Designed a custom 4-layer GPS-module & magnetometer PCB on KiCAD fitting within 260mm by 260mm
- Prototyping a 4-layer flight controller PCB, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261

### Autonomous Clue Detecting Robot

*November 2025*

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane
- Created a Qt5 controller GUI which integrates simulation controls and launch of two ROS node scripts, boosting team productivity by centralizing all controls into a single window which normally takes 4+ separate terminals

## Autonomous Payload Retrieving Robot

*June - August 2025*

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

## SKILLS

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|-------------------|------------------------------------------------------------------------------------------|
| <b>Software</b>   | C/C++, Python, Java, Assembly, VHDL, Linux, Git, CMake                                   |
| <b>Embedded</b>   | ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, ROS, Gazebo, FPGA, I2C, SPI, UART              |
| <b>Electrical</b> | Kicad, Altium, LTSpice, Soldering, Oscilloscope, Electrometer, DMM, SMU, Logic Analyzer  |
| <b>Mechanical</b> | Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper |